

Zimin Liang

PhD Candidate

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🌐 <https://Zim-L.github.io/about>

SUMMARY

PhD in Computer Science at the University of Birmingham, UK. Primary research focuses on **multi-objective combinatorial optimisation** (e.g., subset selection, route planning, scheduling and resource allocation) in **black-box** settings (e.g., simulation), particularly on problem characteristics and what algorithms/operators to design/use under different scenarios (fitness landscapes) and different computational budgets.

Other research interests include behaviour of LLM agents in games (e.g., strategies of different QWen3 models in games), artificial life (e.g., Life-history simulation) and different complex systems.

EDUCATION

2021.9 – 2026.7	University of Birmingham	Ph.D Computer Science
<i>Research: Multi-objective optimisation, Multi-objective Combinatorial Optimisation, Performance analysis, Local Search, Evolutionary Computation, Visualisation of Optimisation Problems.</i>		
<i>Thesis: Multi-Objective Combinatorial Optimisation: When Simple Strategies Meet Complex Problems</i>		
2015.9 – 2021.7	University of Birmingham	MSci Computer Science & MRes Natural Computation (MSci: Integrated Bachelor with Master)
<i>Research: Multi-agent system, Artificial life, Evolutionary Computation, Neural Computation, Computer-aided verification (Formal methods), Chaotic random number generator, Robot programming.</i>		
<i>Thesis: Use Agent-Based Simulation to Investigate the Impact of Social Learning Between (Chimpanzee) Communities</i>		

PUBLICATIONS

1. Z. Liang, M. Li, 2026, "Random is Faster than Systematic in Multi-objective Local Search", In AAAI.
2. R. Ren, Z. Liang, M. Li, C. Qian, 2026, "Not Just for Archiving: Provable Benefits of Reusing the Archive in Evolutionary Multi-objective Optimization", In AAAI.
3. Z. Cui, Z. Liang, M. Li, 2026. "A Problem Landscape Visualisation Method for Multi-Objective Optimisation". In Mathematical and Computational Applications (MCA).
4. Z. Liang, M. Li, 2025, "On the Problem Characteristics of Multi-objective Pseudo-Boolean Functions in Runtime Analysis", In Foundations of Genetic Algorithms (FOGA).
5. Q. Liu, J. Dong, N. Cui, Z. Liang, J. Luo, 2025, "Strategic Allocation of Medical Resources in Disaster Response: A Multi-objective Nonlinear Dynamic Model with Risk and Cost Considerations", In CAIE.
6. S. Ren, Z. Liang, M. Li, C. Qian, 2025, "A Theoretical Perspective on Why Stochastic Population Update Needs an Archive in Evolutionary Multi-objective Optimization", In IJCAI.
7. Z. Cui, Z. Liang, L.M. Pang, H. Ishibuchi, M. Li, 2025, "When to Truncate the Archive? On the Effect of the Truncation Frequency in Multi-Objective Optimisation", In GECCO.
8. Z. Liang, Z. Cui, M. Li, 2024, "Pareto Landscape: Visualising the Landscape of Multi-Objective Optimisation Problems", In Parallel Problem Solving from Nature (PPSN).
9. M. Li, X. Han, X. Chu, Z. Liang, 2024, "Empirical Comparison between MOEAs and Local Search on Multi-Objective Combinatorial Optimisation Problems", In GECCO.
10. Z. Liang, M. Li, P.K. Lehre, 2023, "Non-elitist Evolutionary Multi-objective Optimisation: Proof of Principle Results", In GECCO.

EXPERIENCE

2022 -	Doctorate Associate	Director of Academic Department
Organising academic events: 14 Online academic salons, MSCA Application Workshop, Birmingham Early-Career Researchers Forum AI Sub-Forum 2024 and 2025. Participate in Shenzhen Bao'an Innovation and Entrepreneurship Competition for Overseas Doctorate Entrepreneurs.		
2022 – 2024	University of Birmingham	Teaching Assistant
Modules: Computer Aided Verification (Formal Methods), Evolutionary Computation, Artificial Intelligence 2 (Machine Learning), MSc Final Year Project, Algorithms for Data Science.		
2021 – 2024	University of Jinan	Teaching Assistant
Assisted Dr. Cui Na's research group at the School of Civil Engineering, University of Jinan.		
2018 – 2018	Ark St Alban's	Teaching Assistant
Assisted Mr. Rye's Computing course at year 8 and year 11; Host Scratch coding club for year 4 and year 5.		

SKILLS

- **Programming:** Python, Java; C/C++, CUDA, Haskell, Pascal/Delphi, Visual Basic, Scratch
- **Libraries & Tools:** NumPy, PyTorch, BoTorch, jMetal, pymoo, OpenAI API, SLURM/sbatch, etc
- **Languages:** Mandarin Chinese, Cantonese Chinese, English (IELTS 8)